

1 Basic Settings

Menu	Master data → System settings → Basic settings
Transaction code	setup
Function authorization	setup



Incorrect and/or subsequent changes can seriously affect the complete data collection process.

For this reason, make the below configurations before system implementation.



Most parameter changes only become effective once HYDRA has been restarted.

Purpose

The system-specific configurations in the basic settings define the specifications that apply for the complete system. The basic settings therefore control the processing in the different subsystems (e.g. terminal, posting, etc.).

Integration

A multitude of applications and functions use the specifications defined here. Refer to the documentations of the applications for descriptions of the settings/configurations made for an application.

Field descriptions - General

HYDRA version

Entry of the HYDRA Kernel version. This number is used to inform the customer and can be used to perform HYDRA-internal checks. In general, the software modules work with several versions and can distinguish between the versions.

Kernel version

Entry of the serial number of the HYDRA Kernel within the HYDRA version. This number is used to inform the customer and can be used to process HYDRA-internal inspections. This enables the software modules to detect whether the installed kernel supports specific processing types or not.

Project name

The project name, which is predefined by MPDV, is used to control individual customer-specific processing methods and is entered during implementation. For default systems "HYDRA" will be used as project name.

MW3.x: Customer number

The MPDV customer number is entered here. The modules are licensed on the server and the customer number is used to request the license keys. For this reason

- correctly use the customer number;
- you must not change the customer number.

MW4.0_{pe}: Installation ID

Enter the MPDV installation ID. The modules are licensed on the server and the installation ID is used to request license keys. For this reason:

- correctly use the installation ID
- and you must not change the installation ID.

System number

You use a unique system number for security reasons. The number entered here is printed in front of the badge number when personal badge cards are printed. When clockings are made on a terminal, a validation check is made for this entry. Because of this check, third-party badge cards cannot make postings in this system.



Must be defined by the customer before implementation when existing badge cards are used.



If you use dormakaba terminals, this system number (dormakaba: company number) is checked as mandatory parameter. For this reason, this field must not remain empty because otherwise the badge card cannot be read. If you enter a minus sign "-", this check is skipped. Optionally, you can also prefix the system number by the minus sign and directly store it in the terminal configuration.



For security reasons, always operate a PZE/ZKS system with system number.

Length of personnel no., fill characters for personnel no.

The 'Length of personnel no.' and 'Fill characters for personnel no.' data fields are used to define the conversion of the numeric personnel number from the PZE system into the alphanumeric personnel number in the BDE system.

The only fill character permitted is "0". If "0" is used as fill character, HYDRA will format the personnel number aligned to the right and will adapt it to the length of the personnel number with leading zeros.



You must not change the length of the personnel number offhand, if the system already includes personal data. A change of length will imply that data existing at that point in time in the database will have to be converted explicitly.



When Windows terminals are used, the new dialogs must be created in advance.



You must restart the system after a change.



If you must change the lengths, please contact MPDV Support beforehand.



Must be defined by the customer before implementation.

Length of badge number

The length of the staff badge number is specified. The specified length is used for validation checks.

Change length: If you change the length of the badge number, the system automatically changes the already existing badge numbers to this new length.

If you **reduce** the length of the badge number, an error message is issued if badge numbers exist in the [HR master data](#) or in the [Badges](#) where the places you want to delete are not filled with zeros. Also historical data valid in the past is included in this validation check. The system checks all data and it does not matter whether users have access to this data. The badge numbers are also checked if the customer does not use the access control.



If the length of the badge number must be reduced despite error message, this can only be made by MPDV as part of a custom service that will be charged. The data responsible that the badge number length cannot be reduced must be identified. The handling of this data is then coordinated specifically. Data consistency must always be guaranteed in the process and any badge number (also historical numbers) must be unique in the system at any time. If required, contact MPDV.

The barcode of a staff badge is composed of:

- length of the system number
- length of the badge number (max. 10 digits)
- 1 check digit



You must restart the system after a change.



If you use Windows terminals, you must first create new dialogs.



Must be defined by the customer before implementation.

Length of upload number

If you want to enter OP postings via the upload number on Windows terminals (CTWIN / CTAIP), you can define the field length here. Note the following:

- The length must be identical to the actual upload/confirmation number that is transferred by the PPS system and/or generated in HYDRA.
- If you enter the confirmation/upload number as barcode, it must always have the complete length. Numeric confirmation/upload numbers will not be filled to the length defined in the HYDRA basic parameter settings.



Must be defined by the customer before implementation.

Confirmation/upload number

This option can be used to define whether the confirmation number is generated in HYDRA or not. If the confirmation/upload number is transferred from the PPS system, this option must be set to "Do not generate".

You use the MOC application [Number ranges](#) to define the number ranges.



The number ranges are usually defined by the customer in the PPS. If HYDRA is used as leading system for the generation of number ranges, the customer must define this before implementation.

Length of order no./ operation number

In HYDRA, the posting ID is always the entire order/operation number (the so-called MES order number). This is always the combined number consisting of order number and OP number. If you use sequences and/or splits, the fields "sequence number" or "split number" are added to the "order/operation number".

You can change the length of the order and operation number in specific system areas but they must be constant within the overall system. This configuration must be made before transferring data to the system for the first time. The above configuration specifies at the same time the format of the operation's barcode that is used to make postings for the operations in the system.

Note: It can also be required to make changes in the number range management.

Maximum value: order number: 20

Maximum value: operation number: 10



By default, the number range management provides configurations based on order numbers with a length of eight characters that can be used for the following orders:

- overhead cost or waiting period orders (order type: GKP, GKM), if the order number is generated automatically.
- merged operations (MOP). The system assigns a separate order number to these operations when they are created.

If the order number length deviates, you must also change the length in the number range management.



Must be defined by the customer before implementation.

Length of the sequence number

This input field is only enabled if the relevant license is available .

For SAP customers, HYDRA provides the opportunity to integrate parallel or alternative sequences (condition: this must be possible in the interface). In this case, you have to define which rounding precision shall apply for the sequence number in HYDRA.

Maximum value: 6



Must be defined by the customer before implementation.

Length of split number

This input field is only enabled if the relevant license is available .

In contrast to the previous versions of HYDRA (HYDRA 7 and below), it is no longer the last digit of an operation that is used to specify a split number. Instead, the split number is transferred to a field of its own. The length of the split number is defined here.

Maximum value: 2 (this results in a maximum of 99 split operations per operation; with a split number length of 1, an operation can be split into a maximum of 9 split operations).

The terminal input fields are adjusted accordingly (dynamic dialogs have to be activated once more).

The split number is not shown in the sequencing list on the terminal. As a rule, the split number is not important to the operator. OPs are split for planning reasons. Consequently, the split number only becomes visible, once the OP has been selected from the sequencing list and entered in the input field of the posting dialog. If required, the split number can also be shown at other places on the terminal.



The total length of all fields must not exceed 25 digits. If DOS-based terminals are used, the total length of all fields must not exceed 16 digits.

You must not change the length of numbers offhand, if the system already includes orders. Among others, you must proceed as described below:

- 1) Back up database
- 2) Explicitly convert the data that already exists in the database (MPDV services)
- 3) Change number range management
- 4) If Windows terminals are used: unload (back up) existing dialogs and generate new dynamic dialogs
- 5) Restart HYDRA and all terminals
- 6) Test

You can only do this, if no one is working (no shift), since no orders/ operations must be logged on during the conversion.



If you must change the lengths, please contact MPDV Project Management in good time beforehand.



Must be defined by the customer before implementation.

Currency unit

Use this field to show a currency unit that is displayed in different HYDRA applications.

Please note: The currency will not be processed (e.g. converted).

Time lag for requesting lists

Reloading lists when shifts change strongly charges the system. You can use the "Time lag for requesting lists" option to define the time lag (OFSLST) between terminal groups for reloading lists. This way, a better distribution of the load is possible.

Ten points in time are calculated when the terminals are started. Each of these points in time is delayed by the value entered in the 'Time lag' field. At each of the calculated points in time, a group of terminals starts. If the terminal numbers are sequence numbers, the delay is allocated using the numbers so that the groups have a similar size.

Example:

Time lag: 10 seconds; 35 terminals are connected (terminal numbers 1-35).

The formula $((\text{terminal number} - 1) \% 10) * \text{OFSLST}$ will lead to the following behavior when shifts change:



At the time of a shift change: 1, 11, 21, 31 lists are reloaded

10 seconds later 2, 12, 22, 32 ...

again 10 seconds later: 3, 13, 23, 33 ...

again 10 seconds later: 4, 14, 24, 34 ...

again 10 seconds later: 5, 15, 25, 35 ...

again 10 seconds later: 6, 16, 26 ...

again 10 seconds later: 7, 17, 27 ...

again 10 seconds later: 8, 18, 28 ...

again 10 seconds later: 9, 19, 29 ...

again 10 seconds later: 10, 20, 30 ...

Field descriptions "BDE"

BDE maintenance based on events

This mode activates the function of the tabular event maintenance as editing function. The maintenance of posting function can still be used to evaluate HYDRA postings but corrections will no longer be possible.

If this option is not set, the tabular event maintenance function is disabled, i.e. you can no longer edit, correct and recalculate events. In this case, you can correct entered data via the [Order-related postings](#).

Editing of scrap with reason (partial quantity postings)

This option is no longer evaluated since MES Weaver 2.0. In principle, partial quantity postings (postings of the "T" record type) can now be shown and edited in the [Order-related postings](#).

Generating MDE postings

This option is required to execute MDE evaluations/reports if terminals without machine data collection are used. In this case, MDE postings are generated using the BDE module.

If this option is not set, no MDE postings are generated.

Posting manual quantities as automatic quantities

Use this option to book manual quantities as automatic ones, i.e. the quantities are booked to all operations and persons logged on.

Only merged operations generated on the terminal are excluded. In this case, manual quantities are only booked via specific posting functions for merged operations.

If this option is not set, manual quantities are only booked for the logged-on OP.

This option only affects postings at MDE machines, i.e. the machines assigned to an MDE terminal.



Only activate this option after consultation with MPDV.

Enhanced split function

If you are authorized, you can split operations in the tabular or graphic order sequencing of the BDE module or in the graphic planning board of the HLS module.

Once you have enabled this option, the split function is converted from the ["simple" split function](#) to the ["advanced" split function](#).

Booking production time to MUT during break

This option is used to book production times arising during a break defined in a shift model to the RPA main utilization time (MUT) and as labor utilization. Downtime periods during the break will be booked to the RPA of the break (FP) and be hidden in the labor utilization.

If this option is not set, the break duration defined in the shift model will, in general, be booked to the RPA of the break (FP) and hidden in the labor utilization even when the production was running during this period.

See also the example based on the following configuration.

Proportionate RPA posting in personnel postings

Using this option, the system makes proportionate bookings for the number of parallel logons of a person (number of operations and machines) to the person-related resource performance accounts in the log records of record type U, E and B.

If this option is not set, no proportional RPA booking is made.



This function cannot be used together with the following other settings:

Basic settings → BDE tab → Process merged operations → "Standard time (terminal)" or "Default quantity (terminal)"

[Terminal configuration](#) → MF functions → Generation → "Per machine"

Synchronizing labor times with the person's BDE shift model

The times accumulated in "resource performance accounts", the "labor utilization" in the personnel postings and the "labor utilization" in the order postings are synchronized with the employee's personal shift model.

To do so, store a personal shift model in the [HR master data](#) of the person concerned. If 0 or an inexistent model is stored in the personal shift model, the person's times will be synchronized with the machine's shift model.

If a person is logged on longer than specified in the shift model defined for them, this time is not posted to the resource performance accounts and the labor utilization. The posting times/logon times themselves are not changed.

Also the breaks are identified and posted according to the personal shift model.

Generating log record "finish OP"

If this option is active and you finish the operation via the [Order overview](#) and the function "operation status", a corresponding record (posting with the record type "E") is generated and the operation is set to the status "E". This record is transferred to the higher-level PPS system.

If you use the "change status" function to "finish" the complete order, a final posting is generated for all operations of the order and their status is set to "E".

If this option is not set, the system does not generate a final posting. When you finish an operation or order, only the operation status is set to "E".

Check for PZE clock-in

This option is used to check if a person is present in the PZE system when this person logs on to a machine or when there is an advance logon to a machine. If this the person is not present in the PZE system, the system will refuse the staff logon and/or advance logon to the machine.

N No check

Validation check is inactive

J Check with staff logon

Validation check is only activated for staff logons

M Check with staff logon and advance logon (MDE)

Validation check is activated for staff logons and advance logons

Processing of merged operations

The documentation [Processing of Merged Operations](#) describes the possible options and their processing.

Sequencing list with OP logon

Defines whether a sequencing list based on machines or machine groups is used.

Subject to the production structure and depth of production control, the user uses this configuration to control the selection list of operations that can be logged on.

- M Sequencing list based on machines/workplaces. The sequencing list only shows those OPs that are actually planned (in detail) for the workplace/machine.
- G Specific to groups. The sequencing list shows those OPs that are either planned for the current workplace or another workplace of the group or that are still located in the pool for the group.

You can override this system setting if you select the "sequencing list" option in the workplace configuration.

RPAs to calculate labor times

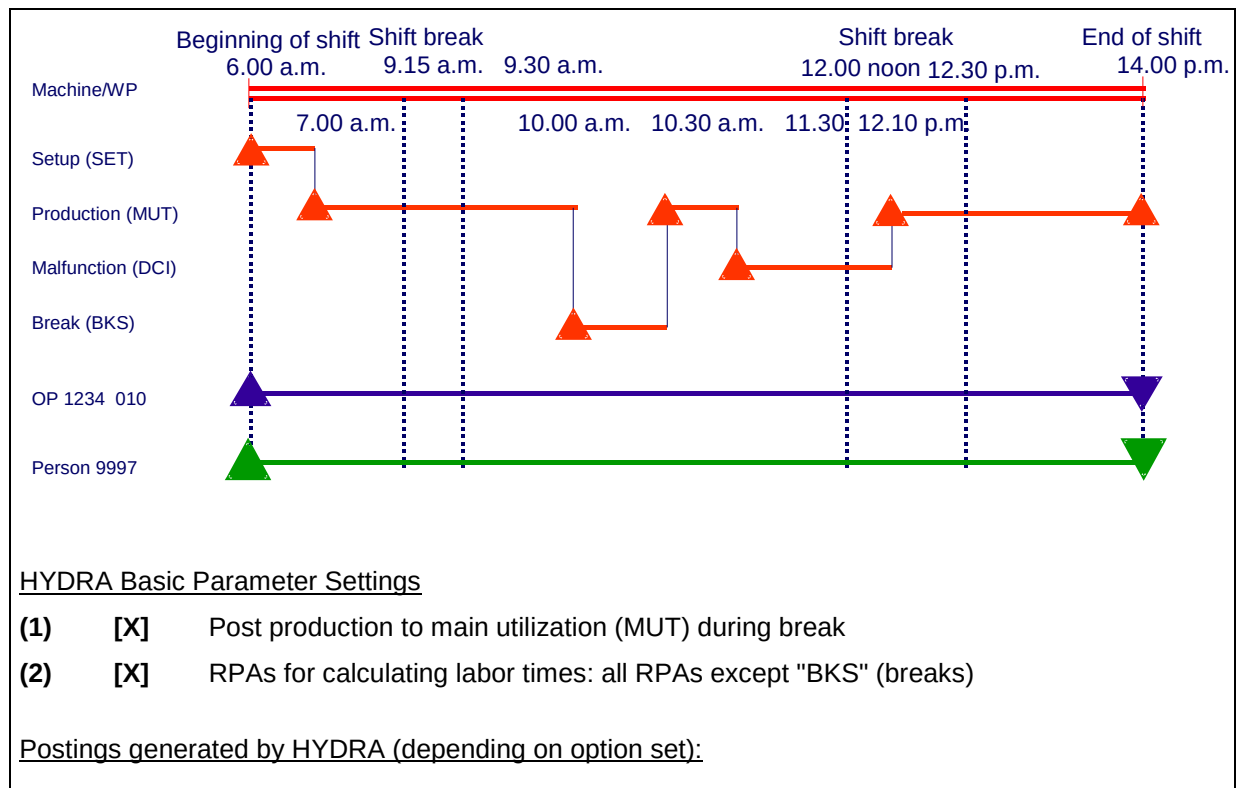
If this option is set, you must specify the times that are used to calculate the labor utilization.

If this option is not set, all times are used for the labor time (e.g. breaks).



This option is used in combination with the option *Post production time to MUT during break*. See the following example

The following example illustrates the two configuration options above. It is assumed that all RPA times except RPA 12 "BKS" (break) are used for the calculation according to option *RPAs for calculating labor time*.



(1) []	(2) []					
RPA SET	1:00 hrs.	RPA DCI	00:30 hrs.	Personnel deployment/labor utilization*	7:15 hrs.	
RPA MUT	5:15 hrs.	RPA BKS	1:15 hrs.			
(1) [X]	(2) []					
RPA SET	1:00 hrs.	RPA DCI	00:30 hrs.	Personnel deployment/labor utilization*	7:50 hrs.	
RPA MUT	5:50 hrs.	RPA BKS	00:40 hrs.			
(1) []	(2) [X]					
RPA SET	1:00 hrs.	RPA DCI	00:30 hrs.	Personnel deployment/labor utilization*	6:45 hrs.	
RPA MUT	5:15 hrs.	RPA BKS	1:15 hrs.			
(1) [X]	(2) [X]					
RPA SET	1:00 hrs.	RPA DCI	00:30 hrs.	Personnel deployment/labor utilization*	7:20 hrs.**	
RPA MUT	5:50 hrs.	RPA BKS	00:40 hrs.			
* Assumption: 1-machine operation If several machines are operated the posting will be proportional ** the following times are deducted: 10:00 - 10:30 AM and 12:00 - 12:10 PM						

Interfacing to the Engel Monitoring System (EMS)

If this option is enabled, Windows terminals can be connected to the Engel Monitoring System (EMS):

- HYDRA terminals can trigger the transfer of CNC data via EMS
- data will then be adopted from the EMS, instead of a machine interface
- Via EMS HYDRA data will directly be shown on machine displays

When interfacing to the Engel Monitoring System is activated, the following additional [Configuration options](#) are available in HYDRA:

PZE (IN/ OUT) controls BDE

Optionally, clocking records pertaining to the Time & Attendance module (PZE) may trigger processing in BDE (Shop Floor Data Collection). You can use the waiting period processing between logoff and logon of persons in BDE; it does not matter if and how it is used in the PZE module. The following options are available for integration with PZE:



"PZE controls BDE" is only processed if the IN/OUT clocking is performed at a PZE terminal. If you create IN/OUT clockings using the MOC application Labor time maintenance, this does not affect the function "PZE controls BDE".

With the functions "PZE controls BDE", a BREAK clocking (beginning of break) is processed like an OUT clocking. The end of break clocking is an IN clocking.

N No control

Postings made in the Time and Attendance do not affect the BDE.

K Waiting period



If this option is in use, you must activate the option "Activate waiting period processing" described below.

When clocking-in, the person is automatically logged on to a waiting period operation in the shop floor data collection system (BDE).

If the user logs on to a production OP during the "waiting period", the duration between the clocking-in and the logon to the production operation is posted to the logged on operation. The logon time of the production operation is postdated to the PZE posting.

The system generates a "waiting period posting" as soon as the waiting period is exceeded. Data posted to the production order is not changed. This processing also applies for the times between the logoff and logon of staff during working time.

The person is automatically logged off with clock-out. If the person was no longer logged on to an operation at the time of the clock-out, a waiting period posting is generated for the time between last logoff of the person and clock-out. This is even the case if the waiting period is not exceeded.

The operation is interrupted automatically, if the last person who works on the operation clocks out (see note below).

A Auto-in



If this option is in use, you must activate the option "Activate waiting period processing" described below.

Waiting period processing (as described in the "waiting period" section above) is enabled. If an operation where the person worked is interrupted automatically, this operation is logged on automatically with the first IN clocking of a person. Requirements for the logon: this operation must not be running or this operation has already been finished.

Example:

If five employees are logged on to an operation and they clock out one after the other, the operation will be interrupted automatically, as soon as the last employee clocks out.

If one of these employees clocks in first the next time, we speak of the "first clock-in of a person". When the employee clocks-in, the operation is logged on again.

The logon takes place at the workplace where the person and the operation were logged on before he or she clocked out.

If it is a single workplace, all persons who are already active are connected to the new operation and the new person is connected to all operations that are already active at this workplace.

G Out

If an employee clocks out, then he or she will automatically be logged off from all operations and orders. If the last person logged on to an operation logs off, then the operation will be interrupted.

The waiting period processing is not active with this mode.



The operation is not interrupted when the last person logs off

- if resources are logged on to the operation or
- if inspection operations are linked to the operation.

In these cases, the logoff of the person via PZE clocking depends on the workplace type or order type:

If it is a group workplace or an overhead cost operation, then the PZE clocking logs off the person. The last active/logged on person *cannot* be logged off automatically because in this case one person must always be logged on to the operation.

It is recommended to log on/off or interrupt the operation via explicit postings

- if resources are logged on to the operation or
- if inspection operations are linked to the operation.

Rounding 1st staff logon

This option defines whether the first person logging on to a workplace on the work day is rounded to the first clock-in of the same day. This processing is only effective if:

- the previous field "PZE (in/out) controls BDE" is set to "waiting period" or "auto in" and
- the clock-in triggers the staff logon. The first clock-in is then prematurely rounded according to the rounding mode rules (PZW). This rounding is usually performed later as part of the work day evaluation. Here, the rounding is performed early to identify the time of the logon to the workplace. This rounding does not affect the actual work day evaluation.

Waiting period processing BDE

Activate waiting period processing

To activate the [personal waiting period processing](#), set this option; and fill the following fields.

[Proceed as described in the link to enable waiting period processing in the system.](#)

Waiting time

Use this field to specify a time frame in minutes if waiting period processing is activated. This field specifies the maximum time that can pass between the clock-in of an employee and the logon to an operation. The field also specifies the maximum time that an employee has between logoff from an operation and logon to the next operation. If this waiting period is exceeded, the system will post the excess time to the stored waiting period operation.

Max. waiting period

The maximum waiting period specifies the maximum duration (time gap) that HYDRA waits before discarding waiting period postings. The system always generates a waiting period posting, if waiting period processing is enabled. At the end of a working day, the waiting period posting starts at the time an employee logs off last and ends with the first logon on the next day if this posting is not interrupted by a clock-out with an active option "PZE controls BDE". Define the maximum waiting period, if you want to suppress this data record spanning two working days of an employee. If you set the maximum waiting period to 0, no waiting period bookings are discarded.

RPA

The waiting period must be posted to an RPA account. Enter an account to do so. Note that you can only enter the number of the account, such as 9 for RPA U8 or 10 for RPA U9. These accounts are free accounts and can be used for waiting periods without affecting the events.

Posting to OP

Combined order/OP number. The waiting period is booked for the specified operation. This way, the individual events can be tracked. You can also define a waiting period OP in the HR master. This OP then takes priority over the waiting period OP defined in the basic parameter settings. The document dealing with [personal waiting period processing](#) describes how to identify waiting period OPs. A posting is only performed for times; quantities, etc. are not posted.



You can only change a waiting period operation, if no person is logged on to this waiting period OP at the time of the change. The personnel overview indicates whether a person is logged on to the currently entered waiting period operation.



Only assign operations of the order type "GKP".



Configure additional settings in the HR master in order to use waiting period processing.

Waiting period processing of machines

Activate waiting period processing

To activate the [machine-related waiting period processing](#), set this option; and fill the following fields.

Lower waiting period/ RPA for unplanned times

Only active, if the configuration for the "waiting period processing of the machine" is enabled:

Times shorter than this waiting period (specified in minutes) are posted to the resource performance account 11 of the production order. This production order will be logged on after this non-productive interval.

The system generates a separate machine-based overhead cost order for non-productive times greater than or equal to this waiting period interval. This overhead cost order posts the time to the resource performance account specified in the configuration (RPA for unplanned times).

If you want to post times shorter than the waiting period to another resource performance account of the production order, define this with respect to machines/workplaces in the [Advanced object configuration](#) as of HYMW Version 8.1.1.512. Use the following parameters:

Object type	BDE
Object ID1	WAITING-PERIOD-PROCESSING
Object ID2	MACHINE-DEFAULT-RPA
Parameter	<workplace number> (8-digit)
Parameter value	<RPA number> (Possible values: 1..12)
Active	☒

Upper waiting period/ RPA for planned times

Only active, if the configuration for the "waiting period processing of the machine" is enabled:

For non-productive times at a machine, use this second configuration to define a time limit (in minutes). When the time limit is then exceeded, this time will be posted to a different RPA (RPA for planned times) as part of the machine-related overhead cost operation. Purpose of this configuration is to transfer unplanned periods at a machine directly to a separate account to minimize in advance the subsequent posting efforts for the foreman and/or work scheduling.

Note: If the time without logged-on operation is shorter than the upper waiting period, the time will be posted to the RPA for unplanned times. But if the time is longer than the upper waiting period, the time without logged on operation will entirely be posted to the RPA for planned times (the time is not distributed to both RPAs).

Posting to OP

The waiting period is posted to the operation specified.



You can only change the waiting period operation, if no machine is logged on to that waiting period OP.



Only assign operations of the order type "GKM".

You can define the waiting period operation with respect to the machine/workplace from HYMW version 8.1.1.512 on. For this purpose, enter the MES order number of the operation including the following parameters in the Advanced object configuration:

Object type	BDE
Object ID1	WAITING-PERIOD-PROCESSING
Object ID2	GKM-OP
Parameter	<workplace number> (8-digit)
Parameter value	<MES order number> (length according to basic parameter settings)
Active	☒

Process with sequencing list

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

Entry of scrap reason

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

Entry of interruption reason

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

Entry of personal quantities

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

Entry of operation-related quantities

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

Manual yield recording

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

Manual scrap recording

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also "terminal configuration").

Entry of staff badges (ADE)

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

Entry of staff badges (MDE)

Use the following configuration options to fill the fields for the terminal-specific dialog control for Windows terminals (see also [Terminal configuration](#)).

MES Link Enabling (MLE) – Order number/ operation sequence/ operation number/ sub-operation number - from / to

In general, SAP interfaces always transfer key fields in their full possible length irrespective of their actual use in MES. Define here which characters of the SAP data fields make up the relevant HYDRA data fields. This simplifies identification and helps you to customize these key fields according to the data formats.

For the following fields define the field positions (from/ to) specifying the values that are integrated in the HYDRA database.



The entered number of digits for the order number/OP must comply with the configured number length for orders/OPs in the BDE tab!

Counting of these digits starts with one, i.e. if the order number is configured from 1 to 8, the system transfers the characters from the first until the last (here: eighth) position included. Note here how SAP proceeds to fill data fields depending on their contents:

- Numeric value: aligned to the right with leading zeros
- Alphanumeric value: aligned to the left with trailing blanks.

SAP split number attached to order number and not to OP

No longer supported.

Confirm HYDRA group as actual work center

If this option is enabled and recorded actual postings are uploaded, the system transfers the HYDRA machine group of the actual machine as the actual work center to SAP.

If this option is not set, the system uploads the HYDRA actual machine as the actual work center to SAP.

Suppress badge number in BDE uploads to SAP

If this option is set, you can prevent the person entering data (the person's badge number) from being uploaded from HYDRA to SAP in the PP time tickets. This might be useful in those instances where data is collected in relation to staff in HYDRA, but the ID cards/badges used in SAP are not the same as in HYDRA.

If this option is not set and HYDRA posts data in relation staff, HYDRA transfers the reporting person's badge number and/or time ticket to SAP.

Suppress scrap in the BDE uploads to SAP

Use this option to prevent the actual scrap collected in HYDRA from being uploaded to SAP. In this case, the scrap quantity is always 0.

By default, the scrap collected in HYDRA is transferred with the time ticket to SAP.

Field descriptions - "HLS"**Planning lead time**

Period of time that the planning function identifies as lead time; result: no assignment is made. The period of time specified is the minimum time that is required to implement a planning.

Scheduling:

Use the factory calendar to identify the EST* (earliest start time).

When planning an OP, use the shift calendar of the first "resource" to cover the central demand (i.e. workplaces in HYDRA).

Planning horizon

Reserved. The planner may specify the planning horizon individually in the [graphic planning board](#).

Simulation horizon

Reserved.

Planning time fence

After an automatic scheduling, the system can automatically fix operations that start within the planning time fence.

Fixed point scheduling

The end date of the production order cannot be met if (despite reduction):

- the earliest end time (EET) is later than the end date of the production order (forward scheduling)
- or
- the latest start time (LST) comes before the start date of the production order (backward scheduling).

If this is the case, the following indicators specify whether the system attempts to comply with the start or end of the order:

"S" = Start

"E" = End.

Default scheduling

If no scheduling direction is defined for an order and scheduling is made in HYDRA, use these indicators to schedule the order:

"V" = forward scheduling

"R" = backward scheduling

Calculation of waiting time

To the extent that scheduling is made in HYDRA, use one of the following options to calculate the wait time of an operation:

G: Calculation of wait time according to the Gregorian calendar (by default)

V: Calculation of wait time according to the shift model of the group (if not yet planned) and/or of the workplace (if already planned) of the preceding operation in the order network.

N: Calculation of wait time according to the shift model of the group (if not yet planned) and/or of the workplace (if already planned) of the succeeding operation in the order network.



The waiting time is only a buffer extending the time distance between neighboring operations by way of scheduling and that thus configures the dates resulting from scheduling. If planning is performed in the HYDRA Shop Floor Scheduling module, the system ignores the waiting time.

Consider priority

Automatic, server-based planning (specific processing) does not include orders with priorities smaller than the value set here. If this field is empty, then planning includes all orders (irrespective of their priority).



Checking is based on the priority that is stored to the order header and not to the one that is defined for the operations.

Displacement to the left

In case of an automatic planning of operations, this option can specify different start dates for the scheduling. Condition: The start of the operation must be within the planning horizon.

If this option is disabled, then the scheduled start date will be the earliest possible date (forward scheduling) or the latest possible start date (backward scheduling).

If the option is enabled, the "left shift" is activated and the scheduled start date is set to "now" plus planning lead time. In other words, an attempt is made to plan the operation as early as possible.



This option ignores the scheduling result, i.e. you can plan the operation before the earliest start date. If the scheduling is performed in the higher-level system, we recommend to disable this option.

Material and Production Logistics

These options are not processed in the HYDRA default configuration.

Field descriptions - "PZE"

Max. interval for clocking supplement

Employees, who forgot to clock out, will automatically be set to "absent" after the time specified here. This ensures with the terminal operation mode "Autom. status" a clock-in will be registered for the employee the next morning.

Another effect of this setting is that in case of a presence exceeding the time specified here, the clocking records will not automatically be comprised to one clocking record. Correct this manually in the "Maintenance of daily results" dialog.

PZE as SAP subsystem/ Version

Use this option to define whether PZE is used as SAP subsystem. If this field is activated, the collected clocking records are transferred to SAP and the daily evaluation will not be started in HYDRA.

The 'Version' field specifies which interface to SAP is used. Enter a value less than 4.5 to transfer the clocking records to SAP via the KK1 interface. Enter a value from 4.5 on to enable the HR-PDC interface.

Process alternate clocking

If this option is activated and a present person clocks in, the system completes the previous clocking by a clock-out. This processing is used for example in those instances where the PZE cost center changes are collected. By default this field is disabled.

Alternate clocking for absence reason

Use this option to specify that depending on a person's status, a clock-in or clock-out will automatically be entered in case the person records an absence reason. By default this field is activated.

Set access statuses (ZKS)

This setting defines that employees are maintained with the following statuses:

- "After entry" after having entered the company premises
- "Before exit" after clocking out. This option is only useful if you control the entrances and exits using Access Control.

Process check digit

This attribute specifies whether the badge number on a barcode badge includes a check digit.

Sign manually created and changed clockings automatically

This option defines whether postings resulting from manually created or changed clocking records shall automatically apply as signed or whether they are subject to approval, if configured correspondingly.

Field descriptions - "PDV"**PDV active (no longer relevant as of PDV 8.3)**

Defines a flag, input options: Yes (J); No (N). Default: J.

Distributor Offset (no longer relevant as of PDV 8.3)

Defines the buffer time of the distributor (in seconds). This option does not calculate the entire period until starting the distributor, but only adds the buffer time specified here, in case data is "still being transferred". Default: 1200 seconds (20 minutes).

Configuration monitor interval

Defines the time interval in seconds, in which the configuration monitor searches for files to be transported. Default: 60.

Number of online tables (no longer relevant as of PDV 8.3)

Defines how many tables are kept online. Default: 10 (10 days online data). Use this option to calculate when a table will be exported. Specify the "Number of online tables" according to the data volume and the customer's hardware and database system.

Transport size (no longer relevant as of PDV 8.3)

Defines the maximum transport size for files (transport function= F) until the PDV.DLL transfers data to the storage component. Unit: bytes, by default: 2000000.

Transport interval (no longer relevant as of PDV 8.3)

Defines the maximum transport time, i.e. the maximum time in seconds after which the current data file must necessarily be closed. Default: 600.

Time lag (no longer relevant as of PDV 8.3)

Defines the maximum time lag (in seconds) between entering changed target values in the server and on the collection level. Default: 10.

Cancellation period (no longer relevant as of PDV 8.3)

Defines the maximum period (in seconds) for recalculating the cancellation. Default: 3600.

Online visualization (no longer relevant as of PDV 8.3)

Defines whether online visualization is available. Possible settings: Yes (J), No (N). Default: N.

Transport function (no longer relevant as of PDV 8.3)

Defines the transport function. This transport function can be exchanged but not by the user. The used function depends on the data volume and does not refer to individual collection components, i.e. it applies globally. Setting options: FILE (F); (future use: STREAM (S), PDM (P); ODBC (O)). Default: F.

Show changed target values (no longer relevant as of PDV 8.3)

This option specifies whether the times when a target value is changed are displayed on the server/terminal (SPC = set point change, target value change). Setting options: Yes (J); No (N). Default: N.

Target medium: measured data (no longer relevant as of PDV 8.3)

Defines the target location and/or storage medium for the measured values. Use one of the two options: store single values to the database system or store single values to the file system. Possible settings: Database (D); file system (F). Default: D.

Archiving path (no longer relevant as of PDV 8.3)

Defines the [path](#), to which exported tables are outsourced.

You can change the path during running operation as the complete path of a generated export file is saved to the archive logs. Default PDVARC --> reference to hy_path. The name of the system directory is optional. Default: \<<MDT>>\pdv\archiv\

Transport path

Defines the [Path](#) for uploading data from the PDV.DLL to the HYPDVSRV. This may be an absolute path that also applies to all drives. Use the masking <<MDT>> to replace the string in the path with the current system directory.

- PDVTRANS: Path to the transport directory, reference to the HYDRA path configuration

Field descriptions - "MPL"

Length of batch number

This value controls the rounding precision (number of characters) of the input field *Batch* in the input dialogs and the MOC evaluations/reports. The system also uses this value to generate the batch number with *Automatic generation of batch number*.

The length must be between 8 and 20 digits.

The batch number length includes the "prefix for automatically generated batch number"!

Example:

If the prefix has 2 digits and you want to have a batch number with 15 digits, then the value for the batch number length is 15. The batch number then has a 2-digit prefix and 13 digits including fixed and dynamic parts. For more details on the structure of the HYDRA batch number, please refer to the document [Batch number generation](#).

Automatic generation of batch no. when creating batches

This option specifies whether HYDRA must create batch numbers automatically when the batches themselves are created manually (e.g. goods receiving batches).

- ☒ Automatic generation of batch no. when creating batches (e.g. create goods receiving batch)
- ☐ The operator has to enter the batch number manually in the input dialog.

Prefix for automatically generated batch numbers

Fixed number of digits in front of the batch number. This prevents an overlapping of number ranges of batches created in external systems and transferred to HYDRA with batches that were automatically generated in HYDRA.

For more details on the structure of the HYDRA batch number, refer to the document [Batch number generation](#).

Create batch	Fixed number of digits in front of the batch number for GR (goods receipt) batches.
Output batch change	Fixed number of digits in front of the batch number for production batches (e.g. for CA_WL).



The number of digits/characters of both values must be identical

Automatic logoff of input batches when logging off the last OP

Option specifying whether all logged-on input batches are logged off, once the last OP logged on to the machine is interrupted or logged off.

Enable this option to log off all logged-on input batches, once the last OP logged on to the machine is interrupted or logged off.

Do not enable the option if you want to prevent the input batches from being logged off automatically.

Creating unknown batches

If this option is activated, HYDRA will collect unknown ERP batch numbers upon the first logon and will also create all required internal entries in the batch management. This means that the ERP batch ID is the only external information in HYDRA.

If this option is disabled, HYDRA will check the ERP batch number entered on the terminal against the ERP batch numbers included in batch management. The system refuses the login, if the ERP batch number does not exist in the batch management.

Field descriptions "ESK"

SMTP server

To send e-mails, enter the IP address or name of the local SMTP server in the *SMTP server* field.

SMTP port

Enter the port of the SMTP server that is used to send e-mails. If a value of "0" is entered here, HYDRA will try to communicate with the server using port "25".

SMTP timeout

Depending on the value entered in this field (in milliseconds), HYDRA will abort a communication attempt when the server does not respond within the defined time span.



There is a time lag in the communication between client and server. For this reason, enter a value greater than 1000ms in field *SMTP Timeout*.

Login type

The field "Login type" specifies if you activate the authentication for the SMTP server or not.

User:

The field "User" is secondary to the field "Login type". If the "Login type" has the value "With authentication", then the fields "User" and "Password" are shown. Enter the user name for the SMTP server in field "User".

Password:

The field "User" is secondary to the field "Login type". If the "Login type" has the value "With authentication", then the fields "User" and "Password" are shown. Enter the password for the SMTP server in field "Password".

SMTP sender

You can enter a central e-mail account. E-mails sent via HYDRA Escalation Management will then have this central account as sender.



At the moment only use the database to edit the plain text description of e-mail addresses shown in mail programs: table: `esk_setup`, column: `.smtp_sendername`.

SMS path

The Escalation Management offers the possibility to send SMS. To do so, store message files in a specific directory. External applications can access this directory to send the messages.

Here, define the [Path](#) where SMS message files should be stored. Use the selection button to access the overview dialog showing the paths that have already been configured.

Pager path

The pager path is configured as the SMS path.